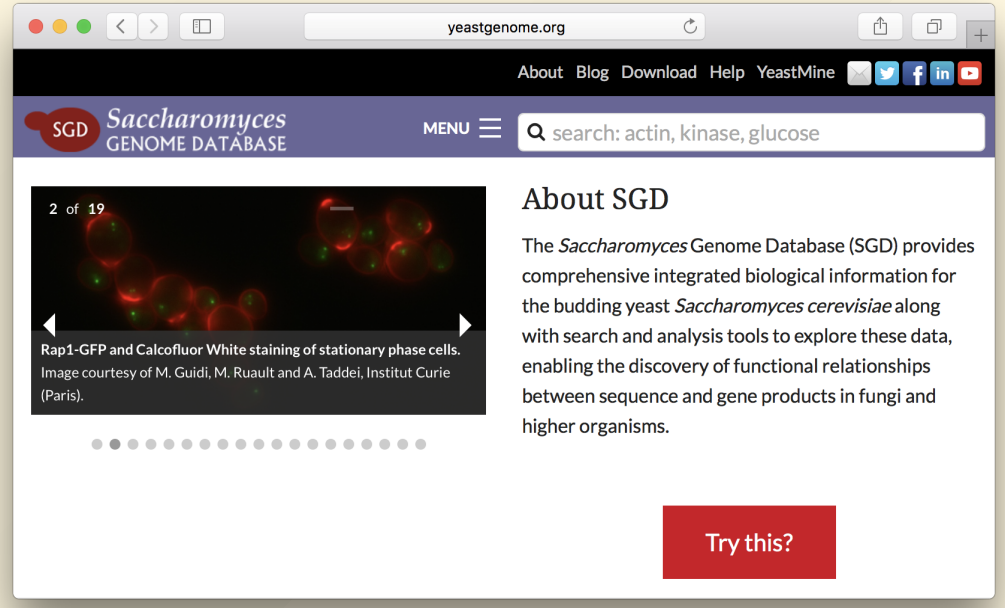
Lecture 3

The Unix Command line for Life Scientists

This presentation tracks the section "Data analytics with Unix" in chapter 4.

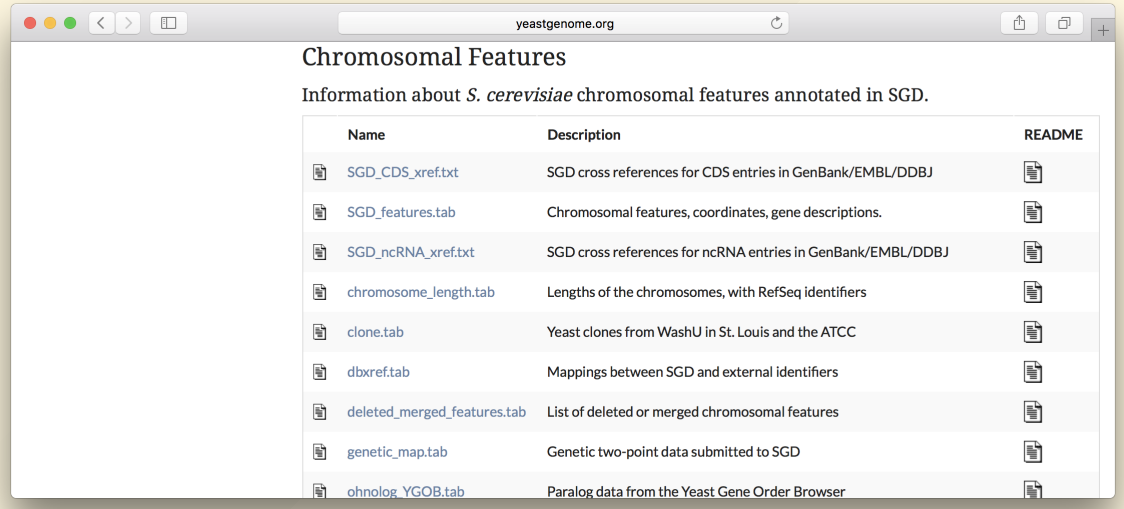
Web based databases



Websites can be handy for exploration. Yet finding specific data may be suprisingly difficult.
How do you find the information on every yeast specific genomic feature at SGD?

Command line is like taking notes

Once you find the content jot it down and use it.



Finally found the URL address (does not quite fit):

```
http://downloads.yeastgenome.org/curation/chromosomal_feat
```

Getting online data

For every lecture create a new folder to work in.

```
mkdir lec03
cd lec03
```

The `wget` command can download a remote file.

```
wget http://downloads.yeastgenome.org/curation/chromosomal
```

Where are you and what have you got here:

```
pwd
ls
```

Paging through files

You can step through a file with the `more` command:

```
more SGD_features.tab
```

keypress commads within `more`:

- `SPACE` or `f` go forward, next page
- `b` go backward, previous page
- `/word` search for a word
- `/` repeats the search for the last word
- `h` help

Think in terms of streams

Instead of treating the file as a UNIT think of it as a stream of information. Open a stream with `cat`

```
cat SGD_features.tab
```

A stream can be made to flow from the left to right from one tool into the next via the pipe `|` character.

Limit the stream by piping into another tool. See what the `head` and `tail` tools do:

```
cat SGD_features.tab | head -1
```

Command line usage tips

Your command line is smarter than Siri! (though that is a very low bar to pass).

1. Recall previous commands with the `ARROW` keys
2. Start writing something then ask your command line to "auto fill" the rest by pressing `TAB`
3. If nothing happens then there are conflicting ways to auto-fill. Double tap `TAB` to see all options

It takes a little practice but you can become very efficient at entering commands.

Simple commands

How many lines, words and characters in this stream:

```
cat SGD_features.tab | wc
```

prints the lines, words and characters:

```
16454  425719 3264490
```

Use flags to customize `wc`. Show the number of lines:

```
cat SGD_features.tab | wc -l
```

A large variety of tools

You don't have to remember them all. The ones you use will stick in your mind. `grep` finds matches in files:

```
cat SGD_features.tab  | grep YAL060W
```

will print (see handbook for unwrapped lines):

```
S000000056      ORF     Verified        YAL060W BDH1    (R
S000036089      CDS                                     YA
```

You can make it color the match with:

```
cat SGD_features.tab | grep YAL060W --color=always
```

Flags can change tools quite a bit

What does not match the word "Dubious" (a weird word that in this file indicates uncertainty about the feature).

Use `grep -v`:

```
cat SGD_features.tab | grep -v Dubious | head -5
```

How many "dubious" and "non-dubious" features:

```
cat SGD_features.tab | grep Dubious | wc -l
cat SGD_features.tab | grep -v Dubious | wc -l
```

How do I store results in a new file?

The `>` character is the "redirection". Instead of the screen it goes into a file

```
cat SGD_features.tab | grep YAL060W > match.tab
```

now check how many files do you have:

```
match.tab
SGD_features.tab
```

You now a subset of the origina data stored in the new file `match.tab`

How do select columns?

It looks like this file uses the feature type (column 2) ORF for protein coding genes. You will need to cut the second field (by tabs).

```
cat SGD_features.tab | cut -f 2 | head
```

prints:

```
ORF
CDS
ORF
CDS
ARS
telomere
telomeric_repeat
```

How do I build my commands

Build your commands one step at a time, always checking that you are on the right track.

Write one command, run through a "limiter" (`head`) then add a new command, rerun it and so on.

Ensure that you understand what the command is doing "so far"

```
cat SGD_features.tab | head
cat SGD_features.tab | cut -f 2 | head
cat SGD_features.tab | cut -f 2 | grep ORF | head
```

Wise man say: "every problem can be solved with some kind of sorting"

Sorting places identical consecutive entries next to one another.

Keep the feature types:

```
cat SGD_features.tab | cut -f 2 > types.txt
```

Sort the feature types:

```
cat types.txt | sort | head
```

Sort + Uniq: a solution to suprsingly many questions

Find unique words. The `uniq` command collapses consecutive identical words into a single entry.

```
cat types.txt | sort | uniq | head
```

How many types of features in this file:

```
cat types.txt | sort | uniq | wc -l
```

It says:

```
44
```

Sort + Uniq

The challenge often is to recognize that the problem can be modeled as a `sort` + `uniq` action.

`uniq -c` also reports the number of times the item has been seen:

```
cat types.txt | sort | uniq -c | head
```

prints the counts and types:

```
 352 ARS
 196 ARS_consensus_sequence
7074 CDS
  50 LTR_retrotransposon
```